Active GLiNER: Which Examples Should LLMs Label?

Abhishek Thomas
Independent researcher, Paris, France
abhi@recolli.com

Abstract

Small extraction models such as GLiNER can be trained on labels produced by a large language model (LLM), which makes annotation cheap but noisy. We ask a simple question: when the LLM can label only a few hundred examples, should the small model choose which ones? We study one round of uncertainty-based selection with a GLiNER2.5 student and LLM teachers on four extraction tasks and six datasets: named entity recognition (CleanCoNLL, BC5CDR, MIT Movie), relation prediction (CrossRE), multi-label classification (Hallmarks of Cancer) and slot extraction to JSON (MASSIVE, English and French). Our pre-registered primary contrast is the interaction between selection and label source: whether choosing the student's least confident examples helps less when an LLM, rather than ground truth, provides the labels. The answer depends on the task. For NER, min selection loses to random under both label sources (min minus random is -6.96 F1 points with LLM labels), mostly because it picks sentences where the student predicts nothing; ranking those last reverses the loss on CleanCoNLL. For relations, min helps with ground truth, and LLM label errors remove most of that gain. LLM errors concentrate where the student is unsure only for NER. For NER, a student trained on 400 LLM-labelled sentences comes close to its teacher (75.28 against 76.36 F1) at 165 times the speed, and on some datasets students beat their teacher. We also trace the thesis mixing curves on every dataset: the student ranks the pool once, takes its least confident sentences, and a share of them gets human labels while the rest keeps LLM labels. From 1,000 ranked sentences, a human share of a quarter or less beats the LLM teacher on six of seven datasets, and LLM labels alone suffice on MIT Movie and MASSIVE; Hallmarks needs three quarters. The result holds for a second LLM family, GPT-6 Luna: a quarter or less beats it on 6 of 7 datasets, and the dev-chosen fewest human labels beat it on test on 7 of 7; a stronger teacher needs somewhat more human labels. The student beats the LLM because LLM labels are free, not because they add accuracy: the same human labels alone do as well, and LLM labels on the whole rest of the pool pull the student back toward its teacher. Replicating the thesis, we also find that its three uncertainty scores (mean confidence, MSE, MNLP) select identical sentences, because every pool holds more empty-prediction sentences than the budget. We release the code as a command-line tool, together with all teacher labels, adapters and run logs.

1 Introduction

Information extraction models that run on a CPU in milliseconds are attractive for production, but they need labelled data for every new domain. Large language models can now produce those labels (Zhang et al., 2023; Kholodna et al., 2024), and a small student trained on LLM labels can match or beat its teacher at a fraction of the cost (Vedula et al., 2024). GLiNER (Zaratiana et al., 2023) and GLiNER2 (Zaratiana et al., 2025) made such students general: one encoder handles entities, relations, classification and structured output from a schema.

If an LLM labels the data, the next question is which data it should label. Active learning answers this for human annotators: pick the examples the current model is least sure about. Yet active learning often fails to beat random selection (Ghose and Nguyen, 2024; Yu and Fathima, 2026; Mittal et al., 2019), and LLM labels change the picture. The examples a student finds hardest may also be the examples an LLM gets wrong, so uncertainty could select exactly the noisiest labels. Hakimi et al. (2026) even find that labelling the whole pool with an LLM beats active learning.

We test this directly. Our question is: *when does*

student uncertainty improve one-round LLM annotation, and when does teacher error erase that benefit? We make five contributions.

1. A pre-registered comparison of uncertainty and random selection, crossed with ground truth and LLM labels, on four tasks and six datasets, with five seeds on primary cells, paired bootstrap intervals and Holm correction.

2. An analysis of teacher errors by student confidence: do LLM mistakes concentrate where the student is unsure (Q2), do different LLMs fail on the same examples (Q3), and do larger teachers label better (Q4)?

3. Mixing curves on every dataset: how many human labels, on top of LLM labels for the student's least confident sentences, it takes to beat the LLM, chosen on dev and confirmed on test (Section 4.1).

4. Cost, latency and size views of students and teachers, including a Pareto front and a quality-latency quadrant chart.

5. An open command-line tool that runs the whole loop (score, select, label with any OpenAI-compatible endpoint, train with LoRA, report), with every paper run reproducible through it.

This paper builds on the author's master thesis, which studied the same loop for NER on MIT Movie with GLiNER and Gemma 3. Section 5 lists what the thesis did, what it found, and which of its protocol problems this study fixes.

2 Related work

Active learning with LLM annotators. LLMaAA (Zhang et al., 2023) uses an LLM as the annotator inside active learning for NER and relation extraction. Kholodna et al. (2024) report that GPT-4 labels in the loop are 42 times cheaper for low-resource NER. ALLabel (Chen et al., 2025) selects demonstrations for the LLM rather than training data for a student. Qi et al. (2026) combine several small LLMs and use their disagreement to flag errors, and HyPAC (Zeng et al., 2026) routes examples between fast and slow LLMs and humans with an error guarantee. Active knowledge distillation (Luccioli et al., 2025) sends uncertain examples to an LLM teacher for classification. We differ in the student (a schema-driven GLiNER2.5 model), the breadth of tasks, and the explicit test of whether teacher error cancels the gain from selection.

Doubts about active learning. Random selection is a strong baseline (Ghose and Nguyen, 2024; Mittal et al., 2019). For NER, Yu and Fathima (2026) find uncertainty and diversity rules unstable, and a scoping review finds that only 13 of 62 studies report run time (Kohl et al., 2024). Practitioners cite setup effort and unclear savings as blockers (Romberg et al., 2025). We therefore pre-register the contrast, report seeds and intervals, and release a tool.

LLM-to-small-model distillation. GLiNER-BioMed (Yazdani et al., 2025) and FiNERweb (Golde et al., 2025) train GLiNER-style students on LLM labels without selection. Vedula et al. (2024) show that a distilled BERT can beat its teacher. Merdjanovska et al. (2026) reweight noisy LLM NER labels by error type.

GLiNER and pre-annotation. The closest work to ours outside arXiv uses zero-shot GLiNER to pre-annotate hiking descriptions for human correction and retrains over three rounds (Tafer et al., 2026); it has no selection rule and no random comparison.

Uncertainty in extraction. Span probabilities can stay informative under domain shift (Zheng et al., 2026); calibration metrics for NER are discussed by Hashimoto et al. (2024). Classic active learning for NER uses sequence-level uncertainty such as MNLP (Shen et al., 2017).

3 Setup

3.1 Tasks and data

Table 1 lists the six datasets. We chose sets that are small (at most about 15k training sentences), carefully annotated, and cover four output structures.

For CrossRE we use an adapted protocol: entity mentions are given as input to every model, and the task is to predict all relation types for every ordered pair of mentions, including pairs with no relation. End-to-end triple extraction was too hard for both students and teachers to support a comparison (pilot micro-F1 below 0.08).

Task	Dataset	Pool	Dev	Test	Labels
NER	CleanCoNLL (Rücker and Akbik, 2023; Tjong Kim Sang and De Meulder, 2003)	13,957	3,233	3,427	4 types
NER	BC5CDR (Li et al., 2016)	5,228	5,330	5,865	chemical, disease
NER	MIT Movie, corrected (Liu et al., 2013; Galileo, 2022)	8,797	978	2,443	12 types
Relations	CrossRE, 6 domains (Bassignana and Plank, 2022)	2,519	300	2,446	17 relation types
Multi-label	Hallmarks of Cancer (Baker et al., 2016)	12,119	1,798	3,547	10 hallmarks
Slots (JSON)	MASSIVE 1.1 en-US and fr-FR (FitzGerald et al., 2022)	11,514	2,033	2,974	55 slot types

Table 1: Datasets. MIT Movie has no official dev split: we hold out a seeded 10% of train. CrossRE pools train and dev and holds out 300 sentences as dev. Documents never cross splits in BC5CDR, CleanCoNLL and Hallmarks. Sentences with no gold label stay in the pool.

3.2 Student

The student is GLiNER2.5-multi (Fastino, 2026) (287M parameters, mDeBERTa-v3-base encoder), loaded from a pinned snapshot, trained with LoRA (Hu et al., 2021) through the `gliner2` trainer. Every run starts from a fresh copy of the base model. Hyperparameters come from one Optuna search per task on the dev set (20 trials, the first being the vendor-recommended recipe), frozen before any main run. Training stops after three dev checks without improvement, and the checkpoint with the best dev micro-F1 is kept. CrossRE is the exception: its dev score moves up and down between checks, so every CrossRE run trains all 1000 steps and keeps its best dev checkpoint. For CrossRE the student scores each ordered mention pair with the checkpoint's pretrained relation scorer, applied to the given mention offsets, so that its zero-shot confidence is meaningful for selection. All runs use deterministic GPU kernels in full fp32 precision without TF32, so a seed fixes the result.

3.3 Teachers

The main teacher is Gemma 4 12B (QAT, 4-bit GGUF) served locally with llama.cpp at temperature 0 with JSON-schema constrained output. One prompt per task gives the label names and, where a dev check showed it helped, one short definition per label from the annotation guidelines. The same prompt is used for every teacher. A second teacher from another model family, OpenAI GPT-6 Luna, is called through the API at temperature 0 with reasoning turned off and the same prompt and JSON schema. It labels every pool, dev and test split, and trains the students of the main-table cells (min and random at the main budget) and of the ranked mixing curves, with 3 seeds. Teacher answers are validated: labels must be allowed and spans must occur exactly in the text, starting and ending on a word boundary of the student's tokeniser. A mention that occurs only inside a longer word (for example the rating "r" inside "rated") is dropped and counted as a teacher error. Teacher labels are cached, so every arm and seed sees the same labels for the same sentence.

3.4 Selection

From a frozen pool, a selector picks N sentences in one round. **Random** samples uniformly. **Min** scores every sentence with the zero-shot student and picks the lowest sentence confidence: the minimum span confidence for NER (no prediction counts as 0), the minimum distance of any label probability to the threshold for classification and relations, and for slots the minimum value confidence or, if nothing is predicted, one minus the highest below-threshold candidate. **Diversity** (NER only) clusters student sentence embeddings with k-means and takes the sentence nearest each centroid. Pool scores and embeddings are computed once and shared, so paired arms select identical sentences. Because "no prediction counts as 0" makes min prefer sentences where the student predicts nothing, a pre-registered sensitivity block on CleanCoNLL ranks such sentences last instead.

3.5 Protocol and statistics

Budgets are $N \in \{100, 400\}$ sentences (CrossRE $\{100, 200\}$). Primary cells (min and random, ground truth and Gemma, at the larger budget) have five seeds, all other cells three. Each seed sets pool tie-breaks, data order and LoRA initialisation. We report strict micro-F1 on the official test set; the test set is used once, for final scores only. Two references bound every cell: a lower bound from four zero-shot GLiNER2 models (small, base and multi from GLiNER2.5, and GLiNER2 large), each with its threshold chosen on dev, and an upper bound from training the student on the whole pool, once with ground truth and once with Gemma labels.

The pre-registered primary contrast, per task, is

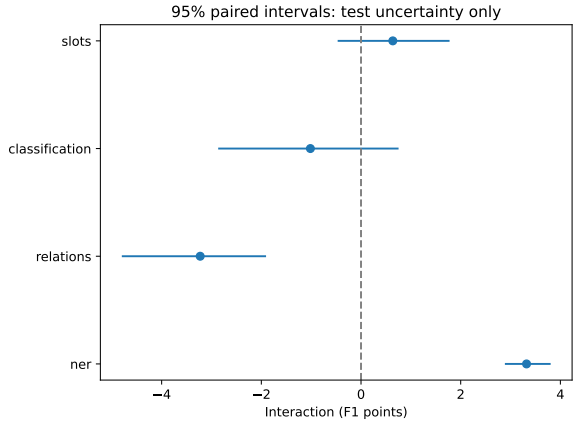


Figure 1: Primary interaction Δ per task with 95% paired bootstrap intervals (test uncertainty only). Positive: min loses less against random with Gemma labels than with ground truth.

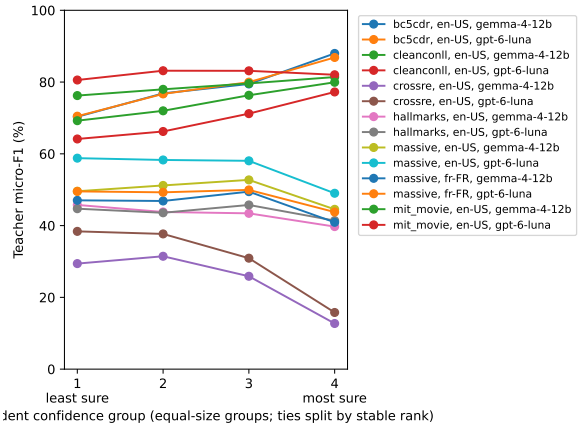


Figure 2: Gemma 4 12B micro-F1 on the pool, by the zero-shot student's sentence confidence (four equal-size groups).

the interaction

$$\Delta = (\text{min} - \text{random})_{\text{Gemma}} - (\text{min} - \text{random})_{\text{ground truth}},$$

paired by seed and, for NER, pooled with equal weight over the three datasets. The four interactions form one Holm family; the four practical contrasts $(\text{min} - \text{random})_{\text{Gemma}}$ form a second. We report seed spread, paired test-set bootstrap intervals (Dror et al., 2018) and ASO (Dror et al., 2019; Ulmer et al., 2022).

4 Results

All numbers below come from 1,887 finished runs through `active-gliner analyse`. Intervals are 95% paired bootstrap intervals over test sentences; they do not include seed variation, so we give the seed standard deviation (SD) of each contrast as well.

Q1: does uncertainty selection help, with ground truth and with LLM labels? Table 2 gives every cell and Figure 1 the primary contrast. The interaction Δ is 3.32 F1 points for NER (95% interval 2.89 to 3.80, Holm-adjusted $p < 0.001$, seed SD 2.93), -3.23 for relations (-4.80 to -1.91, $p < 0.001$, SD 2.97), -1.02 for classification (-2.87 to 0.75, $p = 0.512$) and 0.64 for slots (-0.47 to 1.78, $p = 0.512$). The practical contrast, min minus random with Gemma labels, is -6.96 for NER (-7.34 to -6.63, $p < 0.001$), 1.37 for relations (0.61 to 2.24, $p < 0.001$), -3.37 for classification (-4.48 to -2.32, $p < 0.001$) and 0.36 for slots (-0.37 to 1.10, $p = 0.390$).

The two contrasts must be read together. For NER, the interaction is positive because min loses to random under both label sources and loses less with Gemma labels; there is no gain for teacher error to erase. For relations, the pre-registered pattern appears: with ground truth, min beats random (37.80 against 33.20), and with Gemma labels most of that gain is gone (16.88 against 15.51). For classification, min loses under both sources. For slots, the selectors differ little, and the practical interval includes zero. The NER interaction varies across datasets: its seed SD is close to its mean, and on BC5CDR min loses about as much under both label sources, so the interaction there is near zero (Table 2).

On CleanCoNLL, the no-prediction sensitivity block separates two effects. With ground truth, min scores 70.67, min with empty sentences ranked last 84.46, and random 85.66. With Gemma labels the same three arms score 65.20, 75.07 and 72.24, and training on the whole Gemma-labelled pool scores 73.98. Most of the loss of min therefore comes from the rule that sentences with no prediction count as least confident: min then spends the budget on sentences where the student predicts nothing. With that rule reversed and Gemma labels, 400 selected sentences match or exceed the whole labelled pool on this dataset. This is one dataset and a sensitivity analysis, not a pre-registered contrast.

Q2: are LLM errors where the student is unsure? For NER, yes (Figure 2). Gemma's F1 rises from the student's least sure quarter to its most sure quarter: 70.32 to 87.98 on BC5CDR,

Dataset	Labels	Zero-shot	Random	Min	Min, empty last	Diversity	Whole pool
CleanCoNLL	Ground truth	55.7	85.7 ± 0.9 (5)	70.7 ± 4.1 (5)	84.5 ± 0.6 (3)	86.3 ± 0.9 (3)	89.6
	Gemma 4 12B		72.2 ± 2.0 (5)	65.2 ± 1.7 (5)	75.1 ± 0.3 (3)	70.6 ± 0.4 (3)	74.0
BC5CDR	Ground truth	64.7	82.5 ± 0.5 (5)	72.4 ± 2.2 (5)	83.9 ± 0.3 (3)	82.3 ± 0.5 (3)	87.4
	Gemma 4 12B		75.9 ± 0.7 (5)	65.4 ± 2.6 (5)	76.6 ± 1.8 (3)	75.7 ± 1.6 (3)	79.5
MIT Movie	Ground truth	55.5	84.2 ± 0.2 (5)	78.5 ± 0.7 (5)	84.8 ± 0.2 (3)	84.7 ± 0.7 (3)	88.4
	Gemma 4 12B		77.7 ± 1.7 (5)	74.4 ± 1.3 (5)	78.1 ± 0.5 (3)	78.0 ± 0.3 (3)	79.7
CrossRE	Ground truth	8.4	33.2 ± 2.2 (5)	37.8 ± 0.4 (5)	–	–	41.0
	Gemma 4 12B		15.5 ± 1.4 (5)	16.9 ± 0.4 (5)	–	–	12.5
Hallmarks	Ground truth	13.2	43.0 ± 3.3 (5)	40.6 ± 1.3 (5)	41.2 ± 1.8 (3)	–	42.3
	Gemma 4 12B		38.9 ± 1.5 (5)	35.6 ± 0.8 (5)	37.3 ± 0.8 (3)	–	40.5
MASSIVE en	Ground truth	31.6	70.0 ± 1.1 (5)	69.7 ± 0.7 (5)	70.7 ± 1.0 (3)	–	75.4
	Gemma 4 12B		48.5 ± 1.5 (5)	48.8 ± 1.3 (5)	49.9 ± 0.4 (3)	–	54.1
MASSIVE fr	Ground truth	26.2	62.0 ± 1.0 (3)	64.5 ± 0.9 (3)	64.0 ± 1.0 (3)	–	–
	Gemma 4 12B		46.7 ± 0.7 (3)	47.9 ± 0.4 (3)	47.2 ± 0.6 (3)	–	–

Table 2: Test micro-F1 (%) at the main budget ($N = 400$; CrossRE $N = 200$): mean ± seed SD (number of seeds). Zero-shot is the untrained student with a dev-chosen threshold. Whole pool is one seed. Generated by `active-gliner analyse`.

69.24 to 79.94 on MIT Movie and 76.23 to 81.39 on CleanCoNLL. So uncertainty selection sends the teacher the sentences it labels worst. For the other tasks the pattern reverses at the top: the teacher is weakest where the student is most sure (CrossRE 29.41 against 12.72, Hallmarks 45.76 against 39.72, MASSIVE English 49.57 against 44.54). The bins alone do not predict the sign of the interaction: NER, where teacher errors follow student doubt, has a positive Δ, and relations, where they do not, has a negative one. The level of teacher error matters more: Gemma reaches only 22.62 F1 on CrossRE test, so any selected set carries many wrong labels.

Q3 and Q4: do teachers fail on the same examples, and do larger teachers label better? We compare Gemma 4 12B with the smaller Gemma 4 E4B on frozen, stratified test samples (up to 1,000 sentences, whole documents where the corpus has them), weighting each sentence by its inclusion probability. The larger teacher is better on five of six English datasets: by 6.59 points on BC5CDR (4.15 to 9.10), 8.93 on CleanCoNLL (4.24 to 13.92), 12.85 on MIT Movie (10.45 to 15.26), 4.31 on Hallmarks (0.15 to 8.47) and 13.61 on MASSIVE (10.86 to 16.40). On CrossRE both fail alike (22.39 against 21.82; gap interval -1.85 to 2.97). The two teachers err on largely the same sentences: the Jaccard overlap of their error sentences is between 0.60 (MIT Movie) and 0.70 (CleanCoNLL), and 0.90 on CrossRE. The CleanCoNLL sample holds few documents, so its level estimate (65.16 for 12B) differs from the full test score (76.09); the paired gap uses the same sentences for both teachers and is not affected by this.

A second teacher family. GPT-6 Luna labels better than Gemma on 4 of seven test splits, by 2.61 F1 points on average (Table 3). It gains most on CrossRE (33.34 against 22.62) and loses on MIT Movie (68.78 against 73.95). The two families still fail on many of the same sentences: the Jaccard overlap of their error sentences is between 0.58 and 0.81, highest on CrossRE (0.81). This is a little below the overlap of the two Gemma models on the ladder samples, so many teacher errors come from the task and its annotation conventions, not from one model family. Better labels give only a slightly better student: with min selection, students trained on Luna labels score 0.26 points above students trained on Gemma labels, averaged over the seven datasets. The interaction of Q1 is similar for both families: on the shared seeds, (min − random) with teacher labels minus (min − random) with ground truth averages 0.68 points for Luna and 0.88 for Gemma. These second-teacher numbers are descriptive and outside the Holm families.

Q5 and Q6: better teachers, better students? Can the student beat its teacher? A better teacher gives a better student where we can test it: with 400 random sentences, E4B labels train a student to 68.23 on CleanCoNLL and 35.82 on Hallmarks, against 72.24 and 38.95 with 12B labels (three seeds each). The student can beat its teacher. Trained on the whole Gemma-labelled pool (one seed), it exceeds Gemma's own test F1 on MIT Movie (79.69 against 73.95), BC5CDR (79.54 against 79.04) and MASSIVE English (54.09 against 48.75); with only 400 random sentences it already does so on MIT Movie (77.68) and MAS-

	Teacher F1		Error	Student, Min		Min − Random		
Dataset	Gemma	Luna	overlap	Gemma	Luna	Gemma	Luna	Human
CleanCoNLL	76.1	79.4	0.69	65.5	61.5	−5.6	−12.8	−16.0
BC5CDR	79.0	79.0	0.58	64.2	71.4	−11.5	−4.6	−10.2
MIT Movie	74.0	68.8	0.64	74.8	61.3	−2.9	−10.4	−6.1
CrossRE	22.6	33.3	0.81	16.8	24.6	0.7	4.7	4.2
Hallmarks	42.9	42.0	0.62	35.8	37.3	−3.2	−1.7	−2.4
MASSIVE en	48.7	55.9	0.65	48.5	54.5	−0.4	1.6	0.2
MASSIVE fr	45.6	48.7	0.65	47.9	44.8	1.1	0.1	2.5

Table 3: Two teacher families. Teacher F1 is test micro-F1 (%) of the labels on the full test split. Error overlap is the Jaccard overlap of the sentences each teacher gets wrong. Student columns are test micro-F1 at the main budget ($N = 400$; CrossRE $N = 200$), mean over the 3 seeds that all label sources share. Human is ground truth. Generated by `active-gliner analyse`.

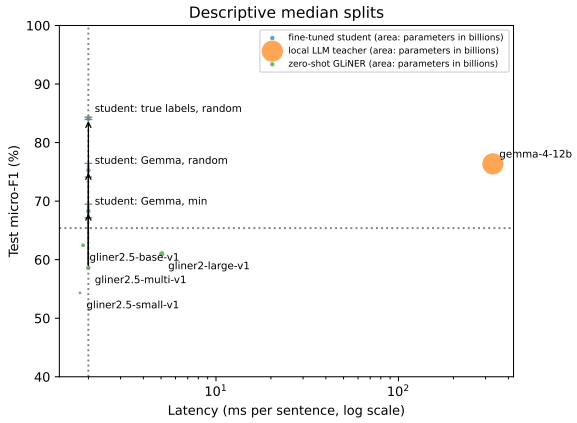


Figure 3: NER, pooled over the three datasets: test micro-F1 against latency on one RTX 3090. Dotted lines are medians. Arrows lead from the zero-shot student to fine-tuned students (400 sentences). Error bars: seed SD.

SIVE French (46.74 against 45.56). On CrossRE it stays far below the teacher (12.54 against 22.62), and the whole-pool CrossRE student trained on Gemma labels scores lower than the one trained on 200 random sentences.

Q7: cost and speed. Figure 3 places students and teacher on one chart for NER, pooled over the three datasets. Gemma 4 12B labels at 329.04 ms per sentence and reaches 76.36 F1. A student trained on 400 random Gemma-labelled sentences reaches 75.28 at 2.00 ms, 165 times faster; the estimated inference cost falls from 20.11 to 0.12 USD per million sentences. The same student trained on 400 ground-truth sentences reaches 84.14, and on 400 Gemma labels picked by min only 68.31; the untrained student scores 58.62. The Pareto front (Figure 6), learning curves (Figure 7) and per-dataset cost curves, which add labelling and selection cost (Figure 10), are in Appendix A.

4.1 How many human labels does it take to beat the LLM?

This is the question the thesis asked on MIT Movie, now on every dataset. The student ranks the pool once, least confident first, with sentences where it predicts nothing ranked last (rule 4 keeps them in the pool, and Section 4.2 shows that ranking them first wastes the budget on them). It takes the top $N \in \{100, 400, 1000, 2500\}$ sentences ($N \leq 1000$ on CrossRE, whose pool holds 2,519). A random share of 0, 25, 50, 75 or 100% of them gets human labels and the rest Gemma 4 12B labels. Three seeds per cell, the same N sentences for every share. Two references bound each panel of Figure 4: the student trained on the whole pool with Gemma labels (lower) and with human labels (upper), both with enough steps for two passes over the pool.

The references. Trained on the whole Gemma-labelled pool, the student lands near its teacher: 74.88 against 76.09 on CleanCoNLL and 78.73 against 79.04 on BC5CDR, and above it on MIT Movie (79.02 against 73.95) and MASSIVE French (50.58 against 45.56). Trained on the whole human-labelled pool, it reaches 89.98 on CleanCoNLL and 88.59 on MIT Movie. The upper reference is not a ceiling: on Hallmarks, 2,500 ranked human-labelled sentences (53.57) score above the whole pool (45.34), whose seeds spread widely.

A small human share beats the LLM. Table 4 gives, for each N, the smallest human share that beats Gemma. The share falls as N grows. On CleanCoNLL, 100 ranked sentences need half human labels (77.13 against 76.09), and 400 need a quarter (79.06). On MIT Movie and MASSIVE, Gemma labels alone are enough once the student

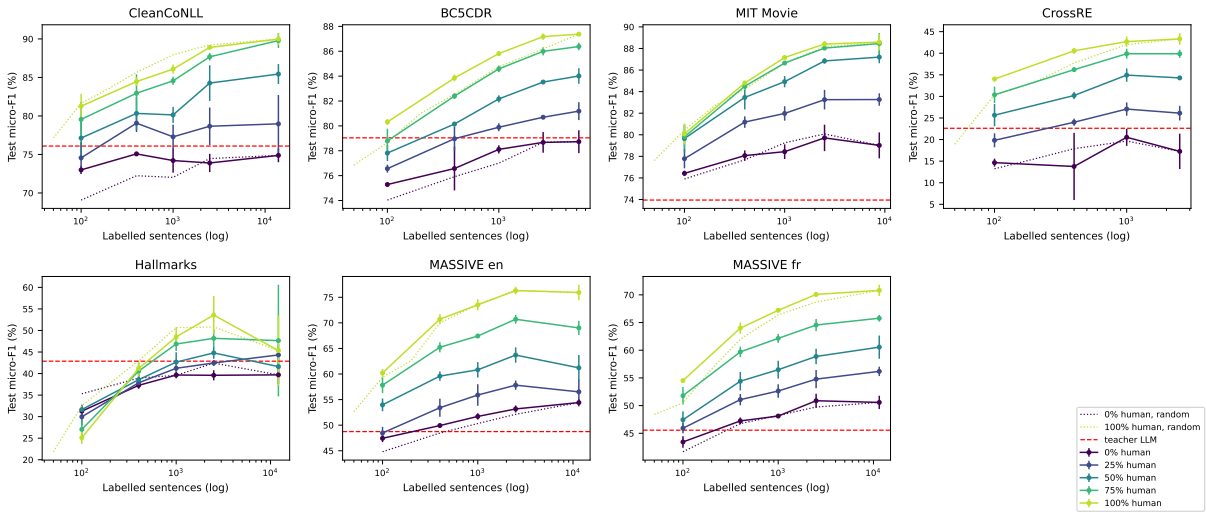


Figure 4: Test micro-F1 against the number of training sentences (log scale), one line per human share; the rest of each set has Gemma 4 12B labels. The rightmost point of each line is the whole pool. Red dashed line: Gemma's own test F1. Dotted lines: random selection with 0% and 100% human labels. Error bars: SD over three seeds.

		Whole pool		Human share to beat the LLM				Fewest (dev)	
Dataset	LLM	LLM	Human	100	400	1,000	2,500	Human+LLM	F1
CleanCoNLL	76.1	74.9	90.0	50%	25%	25%	25%	50+50	77.1
BC5CDR	79.0	78.7	87.4	100%	50%	25%	25%	250+750	79.9
MIT Movie	74.0	79.0	88.6	0%	0%	0%	0%	0+100	76.4
CrossRE	22.6	17.3	43.3	50%	25%	25%	–	75+25	30.4
Hallmarks	42.9	39.7	45.3	none	none	75%	50%	750+250	46.9
MASSIVE en	48.7	54.4	75.9	50%	0%	0%	0%	0+1000	51.7
MASSIVE fr	45.6	50.6	70.8	25%	0%	0%	0%	0+400	47.2

Table 4: Test micro-F1 (%). LLM: Gemma 4 12B on the test set. Whole pool: the student trained on every pool sentence with Gemma or human labels. Human share to beat the LLM: for N ranked sentences, the smallest human share whose mean test F1 is above the LLM (descriptive; "none": no share does, –: not run). Fewest (dev): the cell with the fewest human labels whose seed-mean dev F1 beats Gemma's dev F1 with a paired bootstrap lower bound above zero, as human + Gemma-labelled sentences, and its test F1.

sees enough ranked sentences (MIT Movie 76.42 with 100). Hallmarks is the exception: no share beats Gemma below 1,000 sentences. The last two columns choose on dev instead of reading the test grid: the cell with the fewest human labels whose dev F1 beats Gemma's, then one test comparison per dataset, Holm-corrected over the seven. On test the chosen cell beats Gemma on MIT Movie with no human labels (2.47 points, interval 1.31 to 3.70, $p < 0.001$) and on MASSIVE English with none (2.96, $p < 0.001$), on CrossRE with 75 human labels (7.75, $p < 0.001$), on BC5CDR with 250 (0.85, $p = 0.042$) and on Hallmarks with 750 (3.98, $p < 0.001$). Two dev choices are not confirmed: CleanCoNLL with 50 human labels (1.04, interval -0.33 to 2.50, $p = 0.164$) and MASSIVE French with none (1.67, $p = 0.052$). The dev rule stops at the first cell that clears the bar, which on these two is a cell just above the teacher; larger cells beat Gemma on test by more (Figure 4).

Against a second LLM family. We repeat the ranked curves with GPT-6 Luna as the teacher (Table 5, Figure 9); the human-only cells are shared with the Gemma curves. The result holds. From 1,000 ranked sentences, a human share of a quarter or less beats Luna on 6 of 7 datasets, as it beats Gemma on 6. The dev-chosen cell with the fewest human labels beats Luna on test on 7 of 7 datasets after Holm correction, against 5 of 7 for Gemma. A stronger teacher needs more human labels. On MASSIVE English, where Luna labels better than Gemma (55.93 against 48.75), the fewest cell uses 50 human labels where Gemma needed 0. On CrossRE, where Luna gains most, the dev rule needs 1,000 human labels (9.35 points above Luna, $p < 0.001$). A weaker teacher is beaten without human labels: on MIT Movie, where Luna trails

		Whole pool		Human share to beat the LLM				Fewest (dev)	
Dataset	LLM	LLM	Human	100	400	1,000	2,500	Human+LLM	F1
CleanCoNLL	79.4	76.0	90.0	75%	25%	25%	25%	100+0	81.3
BC5CDR	79.0	79.9	87.4	75%	50%	25%	25%	200+200	79.9
MIT Movie	68.8	71.8	88.6	25%	0%	0%	0%	0+400	72.5
CrossRE	33.3	29.0	43.3	100%	50%	25%	–	1000+0	42.7
Hallmarks	42.0	42.9	45.3	none	none	50%	25%	750+250	45.0
MASSIVE en	55.9	53.6	75.9	50%	25%	25%	50%	50+50	57.9
MASSIVE fr	48.7	48.0	70.8	75%	25%	25%	25%	75+25	51.2

Table 5: The same as Table 4, with GPT-6 Luna as the teacher: LLM is Luna's test F1, the mixed cells have Luna labels for the non-human share, and Fewest (dev) beats Luna's dev F1. The human-only cells are shared with Table 4.

Gemma, 400 Luna-labelled ranked sentences beat Luna by 3.75 points ($p < 0.001$).

Does the ranking matter? Against random selection of the same number of sentences, the ranking gains little on average and varies by dataset. With Gemma labels only, it helps most on CleanCoNLL (73.00 against 69.08 at 100 sentences); with human labels only, it helps on CrossRE (34.02 against 30.35) and hurts on Hallmarks (25.11 against 32.56).

Is the confidence meaningful before training? The ranking only makes sense if the untrained student's confidence says something about its errors. It does, on every dataset (Figure 5, Table 6): the share of correct predictions rises with confidence. Among zero-shot predictions with confidence of 0.9 or more, 74.03% are correct on MIT Movie, 70.21% on BC5CDR and 64.69% on CleanCoNLL, against 20.95% in the lowest well-filled bin of CleanCoNLL (from 0.70). The level differs by task: 11.41% on CrossRE and 31.82% on Hallmarks, where the untrained student is confident and often wrong. Confidence orders sentences well enough to rank them, but it is not a calibrated probability before training.

Gemma labels add little to the same human labels. The mixed cells beat the LLM because Gemma labels are free, not because they add accuracy. With the same human-labelled sentences and nothing else, the student does as well or better: 100 random human-labelled CleanCoNLL sentences alone score 81.64. Adding Gemma labels for the whole rest of the pool to the same 100 human labels lowers this to 74.47 (Figure 8): thousands of teacher labels outvote a few hundred human ones. The pattern holds on all seven datasets: Gemma labels on the rest of the pool help only with very few human labels, 50 on most datasets and up to 200 on Hallmarks (with 50: 40.27 against 21.85), and lower the score beyond that. For a practitioner, the cheapest student that beats the LLM is a few hundred human labels on top-ranked sentences, with or without the LLM.

4.2 Replicating the thesis experiments

We reran the thesis experiments on all seven dataset cells with the protocol above: three seeds, a frozen pool, and dev for every choice.

Thesis selectors measure noise. The thesis compared three sentence scores: mean confidence (avg), mean squared error from full confidence (MSE) and mean negative log-probability (MNLP). In all 63 cells (dataset, label source, budget and seed) the three scores select exactly the same sentences (63 of 63). Each score ranks a sentence with no prediction as least confident, and every pool holds more such sentences than the largest thesis budget: from 563 (6.40%) on MIT Movie to 10,545 (87.01%) on Hallmarks. The selected set is therefore a seeded random draw of empty sentences. For NER, min uses the same rule and selects the same set too (27 of 27 cells). Any score difference between avg, MSE and MNLP is training noise, which makes them a free repeatability test: with the same sentences and seeds, mean F1 differs by up to 4.02 points on Hallmarks, 1.59 on CrossRE and 1.46 on MASSIVE English, but only 0.00 on BC5CDR with ground truth. The thesis ranking of these scores cannot be read as a ranking of methods.

Which weights to train. The default student trains LoRA adapters on the encoder and the task heads. With 400 random ground-truth sentences (three seeds), adapters on the encoder alone score within one point of the default on every dataset (MIT Movie 84.08 against 84.23, Hallmarks 43.67

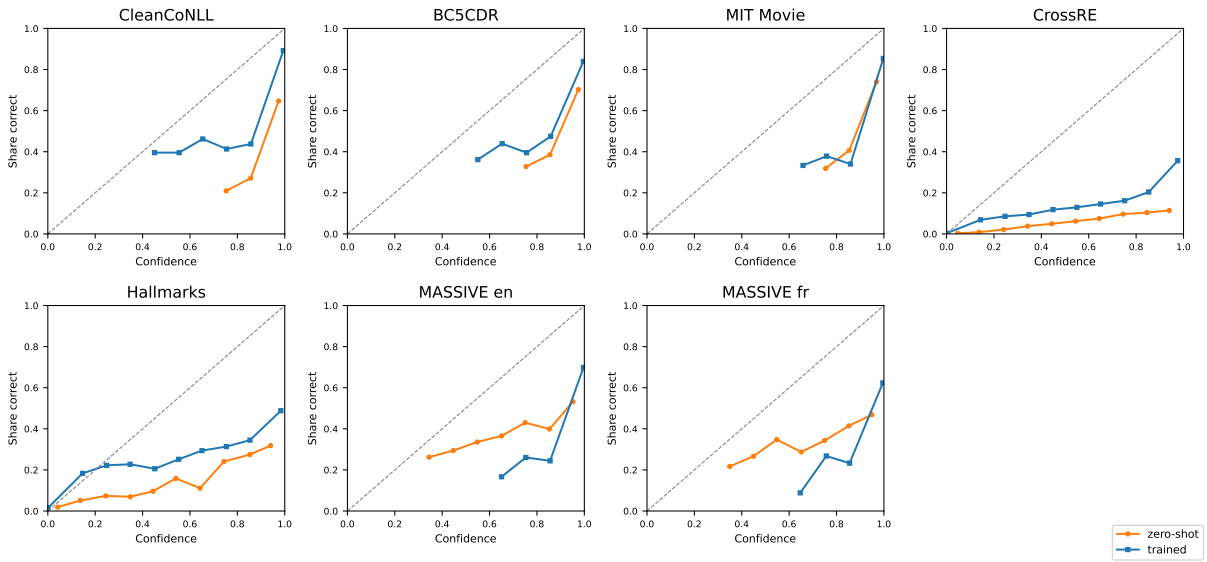


Figure 5: Share of correct predictions against confidence, test set, ten equal-width bins with at least 20 items; extraction counts emitted spans, classification and relations count every label decision. Orange: zero-shot student (the model that ranks the pool). Blue: student trained on 400 random human-labelled sentences (CrossRE 200), seeds pooled. Dashed: perfect calibration.

against 42.98). Training the heads alone loses heavily everywhere (CleanCoNLL 65.14 against 85.66, MASSIVE English 50.07 against 69.98, BC5CDR 73.37 against 82.53) and collapses on Hallmarks (4.13). The encoder carries the adaptation. Full fine-tuning of all weights on the whole ground-truth pool (one seed, learning rate not tuned) scores below LoRA on the whole pool for BC5CDR (85.28 against 87.36), CleanCoNLL (88.79 against 89.63), MIT Movie (87.24 against 88.36) and MASSIVE English (71.88 against 75.36). It is higher only on Hallmarks (51.41 against 42.32), where the LoRA run stopped early (Limitations). LoRA is enough for this student.

Thresholds and budgets. The thesis tuned the decision threshold on test data. On dev, the threshold matters for the zero-shot student and hardly at all after training: dev F1 moves by up to 11.92 points across thresholds 0.3 to 0.7 for the zero-shot student on Hallmarks (11.18 on MASSIVE English), against at most 1.99 for a trained student on any dataset. The learning curves over budgets from 50 to 1,000 sentences are in Figure 7.

5 What the thesis did, and what this study fixes

The thesis ran sixteen experiments on MIT Movie: zero-shot and fully fine-tuned GLiNER, an Optuna search and a LoRA layer study, confidence brackets and a threshold sweep, four selection scores over fourteen budgets, a hard-example comparison of GLiNER and Gemma 3, students on LLM versus ground-truth labels, a mixing grid of ground truth and LLM labels, four teachers, synthetic data, a prompt validator and a cost study. It found, without stating it openly, that Gemma 3's F1 drops from 69.4 on the full test set to 56.9 on the 100 training sentences where GLiNER was least confident; this study turns that observation into question Q2.

The thesis protocol had problems that this study fixes: the MIT Movie dev file was a copy of the test file, so Optuna, early stopping and threshold choice used test data; the pool filter removed sentences with no ground-truth entity; the saved adapter was the last checkpoint rather than the best; the Optuna pruner never received intermediate scores; and no result was repeated with a second seed. Rerun under the fixed protocol on all datasets (Section 4.2), one thesis conclusion does not hold: its three selection scores pick the same sentences, so their ranking was noise. Its central result, that a student trained on a mix of human and LLM labels beats the LLM with few human labels, holds on six of seven datasets (Section 4.1), with the caveat that the same human labels alone do about as well.

6 Conclusion: which examples should LLMs label?

Not the ones the student is least sure about. For NER those are the sentences the LLM labels worst (Q2), and choosing them gains little over random selection once sentences with no prediction are ranked last: with Gemma labels on 400 sentences, CleanCoNLL reaches 75.07 against 72.24 for random and MIT Movie 78.07 against 77.68. The useful split runs the other way. The student ranks the pool once and takes its least confident sentences; a small random share of them gets human labels and the LLM labels the rest. From 1,000 ranked sentences, a quarter or less human labels beats the LLM on 6 of 7 datasets with Gemma and on 6 of 7 with GPT-6 Luna, and the cell with the fewest human labels, chosen on dev, holds on test on 5 and 7 datasets. A stronger LLM raises the bar and needs more human labels. LLM labels matter most when human labels are almost absent: beyond about 50 human labels (200 on Hallmarks), LLM labels on the rest of the pool no longer help (Figure 8). The LLM is a free start, not a substitute for a few hundred human labels on the right sentences.

7 Limitations

Our results cover one round of selection, one student family and English plus French data. CrossRE uses given entity mentions, so its results do not cover end-to-end relation extraction. The hyphen-free tokenisation of the datasets hides a GLiNER2 word-splitting issue that matters for raw biomedical text. We cannot rule out that our models saw the evaluation data. No model owner states that our six datasets were in training, and none states that they were removed. The GLiNER2 family reports zero-shot results on CrossNER, whose sentences and splits CrossRE reuses, so CrossRE test sentences were part of the student developers' entity evaluation; whether checkpoints were chosen on them is unknown. The student's training text includes news and PubMed, so the text (not the labels) of CleanCoNLL, BC5CDR and Hallmarks may overlap. CoNLL-2003 is a known contamination case for earlier LLMs, and public instruction collections contain CoNLL and BC5CDR tasks; our teachers do not document whether such data was used. MASSIVE postdates the student encoder's pre-training text. The full audit is in Appendix B. Training is not exactly repeatable on our GPU. NER runs on ground truth repeat almost to the digit (identical selections differ by at most 0.01 points), but relation, classification and slot runs, and NER runs on Gemma labels, can differ by up to 4.02 points on the same sentences and seed (Section 4.2). We therefore report the seed SD of every cell and of every contrast. Costs are estimates: GPU time at one rental price on one date, and ground-truth labels at a simulated crowd rate of 0.50 USD per sentence taken from a span-annotation study (Kasner et al., 2026). The planned API teachers (Qwen and gpt-oss on Cerebras) were not run; GPT-6 Luna replaces them as the second family, on the main-table cells and the ranked mixing curves with 3 seeds; the pre-registered Holm-tested interaction contrasts and the minimum-human-labels curve use Gemma alone. The smaller E4B teacher labelled only the ladder samples and two training cells.

Three further points qualify single cells. The whole-pool Hallmarks run stopped early on a dev plateau, so it understates what the full pool can give; sentence-level Hallmarks is also much harder than the abstract-level task on which published scores are reported. Six CrossRE runs stopped with a non-finite gradient and one MASSIVE run made no optimiser step; all ran again with the same configuration and seed and finished normally. Runs killed by a GPU fault or out of memory were restarted in a new process with the same configuration and seed. The GLiNER2 trainer silently skips a batch that runs out of GPU memory; a check of epoch against step count in every training log found five runs with skipped batches. We made out of memory stop the run, and those five runs were repeated in full. Finally, fine-tuned students were not timed separately: they reuse the zero-shot timing of their base model, on the unmeasured assumption that a LoRA adapter adds little inference time.

Ethics and broader impact

All datasets are public and used for research as intended by their licences. We release teacher labels as sentence ids and character offsets, never text whose licence forbids it. No human annotators were involved. The GPU time of every run is in the released run logs.

Use of AI assistants

The first version of the code was written by an AI coding assistant (OpenAI Codex) and reviewed by a second assistant (Anthropic Claude), which wrote the later code, fixes and analysis, helped plan experiments and drafted text. The author checked the design, the results and every citation, and takes responsibility for the content.

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A Additional figures

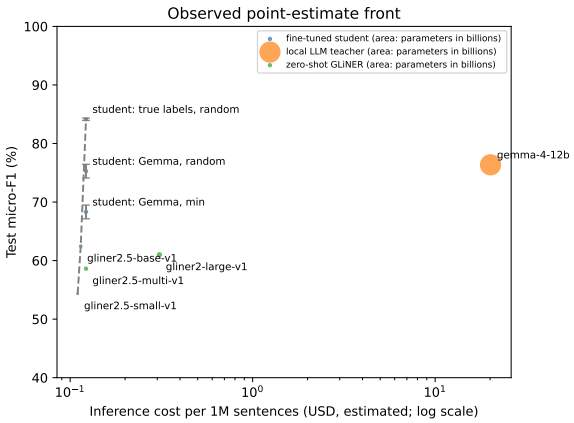


Figure 6: NER, pooled over three datasets: test micro-F1 against estimated inference cost per million sentences (GPU time at one rental price). The dashed line joins the observed points that no other point beats on both axes.

B Data overlap audit

We checked papers, model cards and technical reports (September 2026) for each pair of evaluation dataset and model. We ran no probe, so nothing in this section is measured. Table 7 gives the result. No model owner states that any of our datasets was in training, and none states that it was removed.

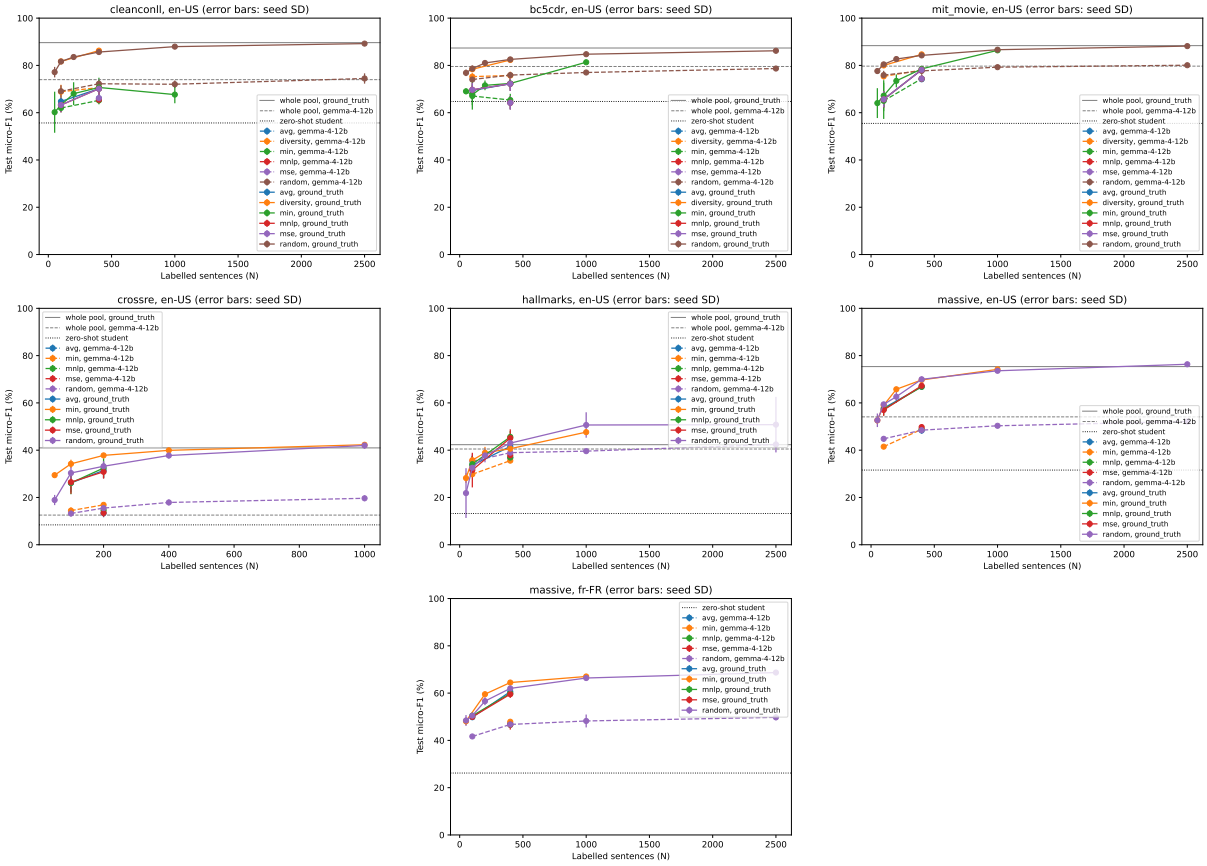


Figure 7: Test micro-F1 against the number of labelled sentences. Solid lines: ground truth; dashed lines: Gemma 4 12B labels. Grey lines: whole pool; dotted line: zero-shot student. Error bars: seed SD.

Dataset	0.5	0.6	0.7	0.8	0.9
CleanCoNLL	–	–	21	27	65
BC5CDR	–	–	33	39	70
MIT Movie	–	–	32	41	74
CrossRE	6	7	10	10	11
Hallmarks	16	11	24	28	32
MASSIVE en	34	37	43	40	53
MASSIVE fr	35	29	34	41	47

Table 6: Zero-shot student: share of correct predictions (%) by confidence bin on the test set; – marks bins with fewer than 20 items.

Dataset	Student	Encoder	Teachers
CleanCoNLL	U	U	P, high
BC5CDR	U	P	P
MIT Movie	U	U	P, low
CrossRE	E	P	P, high
Hallmarks	U	P	P
MASSIVE	U	N	P

Table 7: Possible overlap of evaluation data with model training data. E: evaluated on by the model owner; P: a named path exists but no source confirms it; N: not possible by date; U: unknown. Student: GLiNER2.5-multi; encoder: mDeBERTa-v3 (CC-100); teachers: Gemma 4 E4B and 12B.

Student. The GLiNER2.5 release states that its training data was generated by an agent rather than assembled from existing datasets; it names no teacher, corpus or size and makes no decontamination statement (Fastino, 2026). It reports zero-shot results on five CrossNER domains. CrossRE annotates relations on the CrossNER sentences with the same splits (Bassignana and Plank, 2022), so the student's developers have scored entities on our CrossRE test sentences; whether they chose checkpoints on those scores is unknown. GLiNER2 trains on news, Wikipedia and PubMed text labelled by GPT-4o (Zaratiana et al., 2025), so the text, not the labels, of BC5CDR, Hallmarks and CrossRE may overlap with the zero-shot ladder models.

Encoder. mDeBERTa-v3 is pre-trained on CC-100, built from 2018 Common Crawl snapshots. MASSIVE (2022) and its source SLURP (2020) cannot be in it. Wikipedia and PubMed pages can be, so CrossRE, BC5CDR and Hallmarks text may

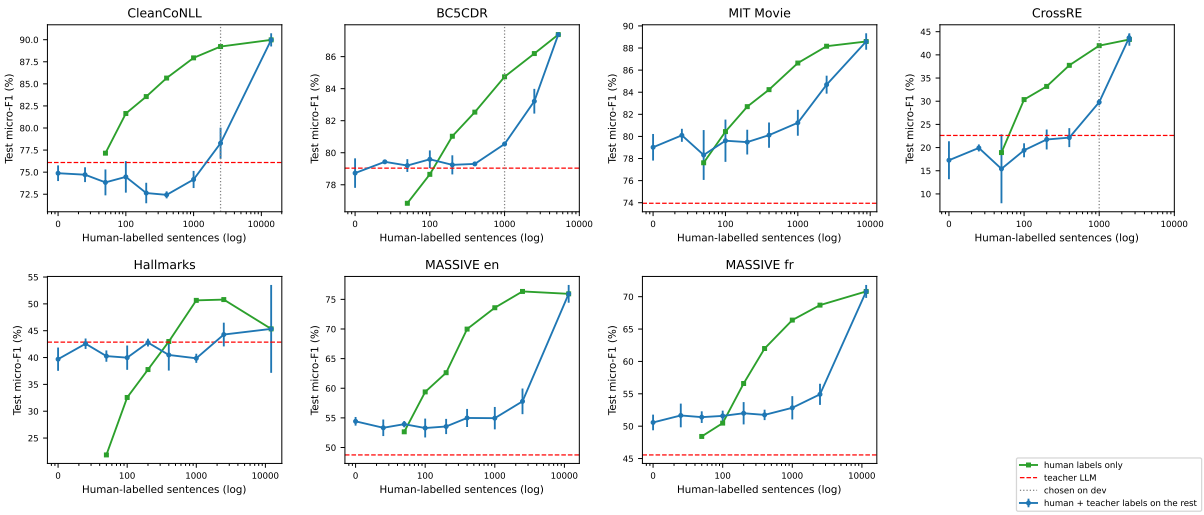


Figure 8: Same human labels, with and without Gemma labels on the rest of the pool. Blue: the whole pool, H sentences with human labels (a nested random set) and Gemma labels on all others; $H = 0$ is drawn at the left edge. Green: the same number of random human-labelled sentences alone. Red dashed line: Gemma's test F1. Error bars: SD over three seeds.

be familiar to the encoder; a masked encoder sees text without our labels.

Teachers. Every dataset was public before the Gemma 4 data cutoff (January 2025). The Gemma 4 report states that benchmarks were decontaminated but does not list them. OpenAI does not publish the training data or a decontamination list for GPT-6 Luna, and all seven test splits predate it. CoNLL-2003 is a documented contamination case for ChatGPT, which reproduces its test sentences with labels. Public instruction collections contain CoNLL and BC5CDR tasks; no teacher owner says whether it used them. We therefore treat CleanCoNLL and the news domain of CrossRE, which is CoNLL-2003 text, as likely seen by the teachers.

Effect on our claims. Exposure raises absolute teacher and student scores. Our main claims compare selection arms, and teachers, on the same sentences. Exposure that is equal across arms does not bias these paired contrasts, but it can narrow the measured gap between teacher labels and ground truth. MASSIVE English and French are translations of the same utterances, so the two locales are not independent evidence.

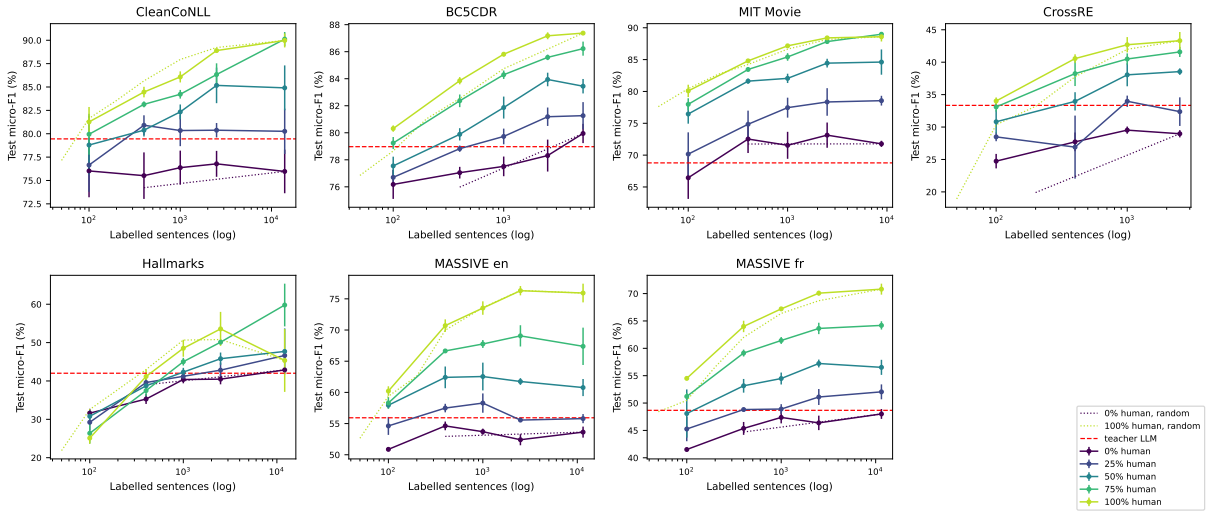


Figure 9: The mixing curves of Figure 4 with GPT-6 Luna as the teacher: the non-human share of each set has Luna labels. Red dashed line: Luna's own test F1. Dotted lines: random selection with 0% (Luna, main budget only) and 100% human labels. Error bars: SD over three seeds.

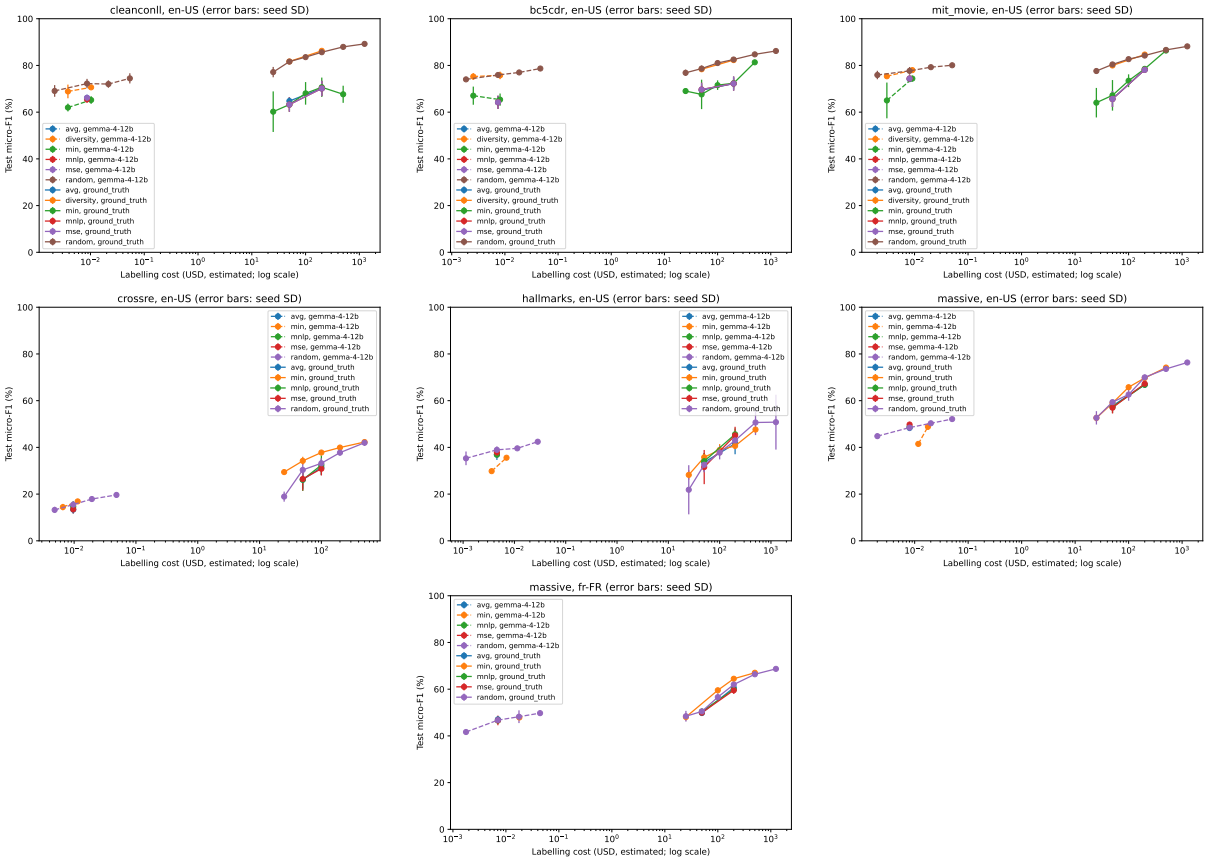


Figure 10: Test micro-F1 against estimated labelling cost: ground truth at a simulated 0.50 USD per sentence, Gemma labels at free-GPU time, plus pool scoring for min and diversity.